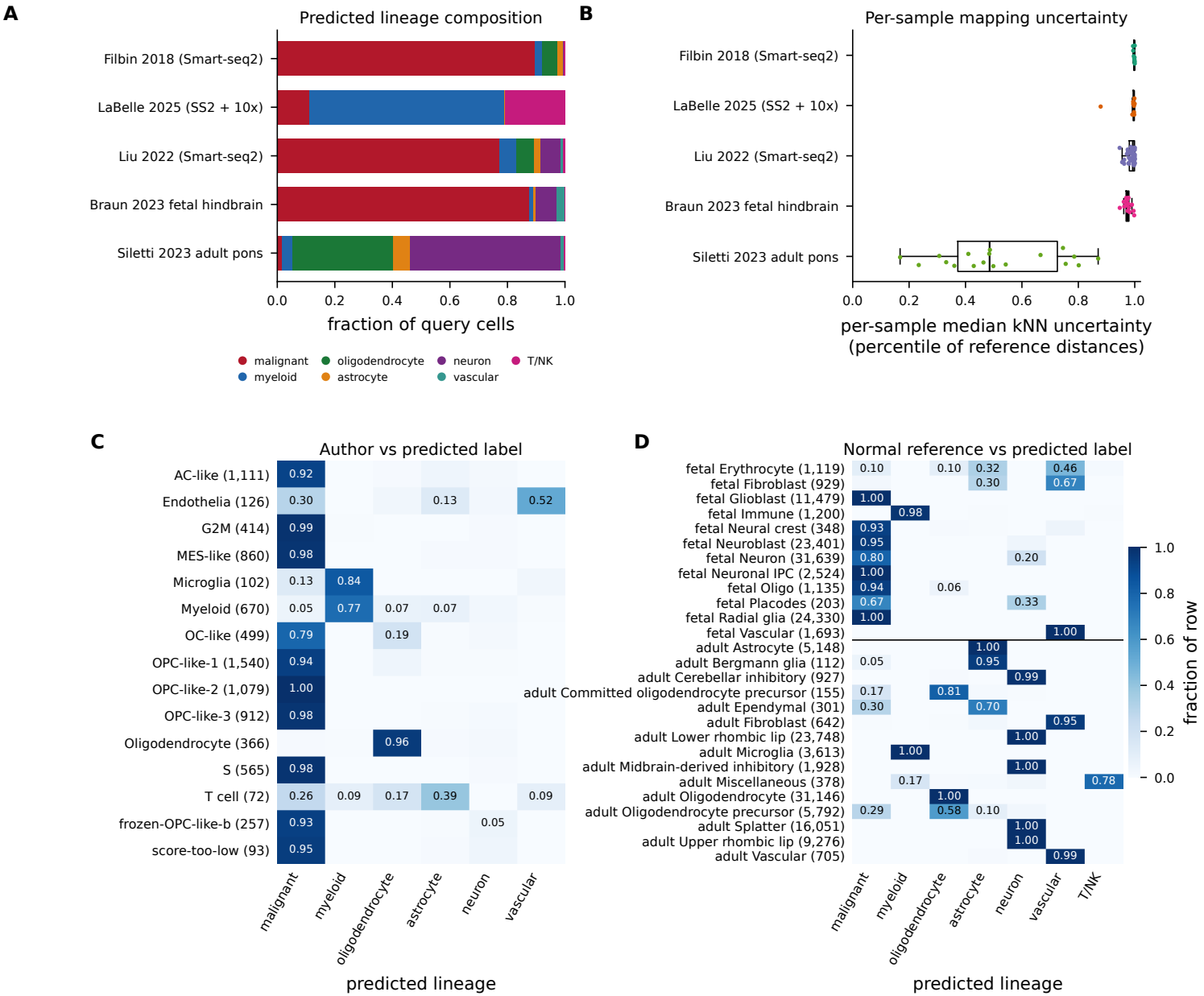
A
Predicted lineage composition
Filbin 2018 (Smart-seq2)
LaBelle 2025 (SS2 + 10x)
Liu 2022 (Smart-seq2)
Braun 2023 fetal hindbrain
Siletti 2023 adult pons
0.0
0.2
0.4
0.6
0.8
1.0
fraction of query cells
malignant
oligodendrocyte
neuron
T/NK
myeloid
astrocyte
vascular
B
Per-sample mapping uncertainty
Filbin 2018 (Smart-seq2)
LaBelle 2025 (SS2 + 10x)
Liu 2022 (Smart-seq2)
Braun 2023 fetal hindbrain
Siletti 2023 adult pons
0.0
0.2
0.4
0.6
0.8
1.0
per-sample median kNN uncertainty
(percentile of reference distances)
C
Author vs predicted label
AC-like (1,111) 0.92
Endothelia (126) 0.30 0.13 0.52
G2M (414) 0.99
MES-like (860) 0.98
Microglia (102) 0.13 0.84
Myeloid (670) 0.05 0.77 0.07 0.07
OC-like (499) 0.79 0.19
OPC-like-1 (1,540) 0.94
OPC-like-2 (1,079) 1.00
OPC-like-3 (912) 0.98
Oligodendrocyte (366) 0.96
S (565) 0.98
T cell (72) 0.26 0.09 0.17 0.39 0.09
frozen-OPC-like-b (257) 0.93 0.05
score-too-low (93) 0.95
malignant
myeloid
oligodendrocyte
astrocyte
neuron
vascular
predicted lineage
D
Normal reference vs predicted label
fetal Erythrocyte (1,119) 0.10 0.10 0.32 0.46
fetal Fibroblast (929) 0.30 0.67
fetal Glioblast (11,479) 1.00
fetal Immune (1,200) 0.98
fetal Neural crest (348) 0.93
fetal Neuroblast (23,401) 0.95
fetal Neuron (31,639) 0.80 0.20
fetal Neuronal IPC (2,524) 1.00
fetal Oligo (1,135) 0.94 0.06
fetal Placodes (203) 0.67 0.33
fetal Radial glia (24,330) 1.00
fetal Vascular (1,693) 1.00
adult Astrocyte (5,148) 1.00
adult Bergmann glia (112) 0.05 0.95
adult Cerebellar inhibitory (927) 0.99
adult Committed oligodendrocyte precursor (155) 0.17 0.81
adult Ependymal (301) 0.30 0.70
adult Fibroblast (642) 0.95
adult Lower rhombic lip (23,748) 1.00
adult Microglia (3,613) 1.00
adult Midbrain-derived inhibitory (1,928) 1.00
adult Miscellaneous (378) 0.17 0.78
adult Oligodendrocyte (31,146) 1.00
adult Oligodendrocyte precursor (5,792) 0.29 0.58 0.10
adult Splatter (16,051) 1.00
adult Upper rhombic lip (9,276) 1.00
adult Vascular (705) 0.99
malignant
myeloid
oligodendrocyte
astrocyte
neuron
vascular
T/NK
predicted lineage
fraction of row
1.0
0.8
0.6
0.4
0.2
0.0